

General Loss Functions & Maximum Likelihood Estimation (MLE)

1. **Q:** What is the primary role of a **loss function** in training Machine Learning (ML) models?
A: It quantitatively measures the deviation between the predicted and true values, guiding model optimization.
 2. **Q:** What is the statistical principle that relates minimizing a loss function to maximizing a probability function?
A: Maximum Likelihood Estimation (MLE) connects loss minimization to maximizing the likelihood of observed data.
 3. **Q:** The minimization of the **Mean Squared Error (MSE)** loss function is equivalent to obtaining the Maximum Likelihood Estimation (MLE) parameters under what specific **noise assumption**?
A: Gaussian (Normal) noise distribution.
 4. **Q:** What mathematical property of the **Mean Squared Error (MSE)** function for linear models guarantees that the local minimum is also the global minimum?
A: Convexity.
 5. **Q:** In the context of the loss function, what does the term **RMSE** stand for, and what unit does it share?
A: Root Mean Squared Error; it shares the same unit as the target variable.
 6. **Q:** For a given loss function $L(\vec{\beta})$, what two conditions must be met (in terms of the gradient and Hessian) for a point $\vec{\beta}$ to be considered a **local minimum**?
A: The gradient $\nabla L(\vec{\beta})$ must be zero, and the Hessian matrix H must be positive definite.
 7. **Q:** If the loss function is continuous and twice-differentiable, what mathematical property does its Hessian matrix exhibit?
A: Symmetry.
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Regression-Specific Loss Functions

8. **Q:** What are two alternative names for the **Mean Squared Error (MSE)**?
A: Quadratic Loss and L2 Loss.

9. **Q:** Why is the MSE considered **sensitive to the presence of outliers**?
A: Because it penalizes larger errors quadratically, giving outliers a disproportionate influence.
10. **Q:** What is the alternative name for the **Mean Absolute Error (MAE)**, and why is it considered more robust to outliers than MSE?
A: L1 Loss; it is more robust because it penalizes errors linearly rather than quadratically.
11. **Q:** What is the main mathematical disadvantage of **MAE** (L1 Loss) for gradient-based optimization?
A: The derivative is discontinuous at zero, complicating gradient computation.
12. **Q:** What specific characteristic of the target variable makes the **Mean-Squared Logarithmic Error (MSLE)** most applicable?
A: Targets that vary exponentially or span several orders of magnitude.
13. **Q:** In the MSLE, how are **underestimations** treated compared to **overestimations** due to the logarithmic scale?
A: Underestimations are penalized more heavily than overestimations.
14. **Q:** What is the purpose of adding "+1" to both the true value (y_i) and predicted value ($\hat{y}_i$) in the MSLE formula?
A: To prevent taking the logarithm of zero and allow all non-negative values.
15. **Q:** What kind of deviation (absolute or relative) does the **Mean Absolute Percentage Error (MAPE)** use?
A: Relative deviation, expressed as a percentage of the true value.
16. **Q:** For what specific characteristic of data is the **Tweedie Loss Function** primarily used?
A: Skewed data with a mix of very low and continuous positive values.
17. **Q:** What are the specific loss functions that the Tweedie parameter ϑ corresponds to when $\vartheta=0$ and $\vartheta=2$?
A: $\vartheta=0 \rightarrow$ Mean Squared Error; $\vartheta=2 \rightarrow$ Gamma deviance.
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Cross-Entropy & Other Classification Loss Functions

18. **Q:** What are the two common alternative names for the **Logistic Loss** function?
A: Log Loss and Binary Cross-Entropy Loss.
19. **Q:** Which specific ML problem type and algorithm primarily utilize Logistic Loss?
A: Binary classification with logistic regression.
20. **Q:** Logistic Loss naturally arises from the Maximum Likelihood Estimation (MLE) of which specific **probability distribution**?
A: Bernoulli distribution.
21. **Q:** For what type of ML problem is the **Softmax Loss** (Categorical Cross-Entropy) primarily employed?
A: Multiclass classification problems.
22. **Q:** What happens to the **Softmax Loss** formula when applied to a binary classification problem?
A: It reduces to the binary cross-entropy (Logistic Loss) formula.
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Analytical Solutions and Numerical Optimization

23. **Q:** Write the final analytical solution for the optimal parameter $\hat{\beta}$ that minimizes L_{MSE} for a simple linear model in the **matrix form**.
A: $\hat{\beta} = (X^T X)^{-1} X^T Y$
24. **Q:** For a quadratic regression model $\hat{y}(x) = \beta_0 + \beta_1 x + \beta_2 x^2$, minimizing the MSE requires solving what type of mathematical system for the parameters β_0 , β_1 , and β_2 ?
A: A system of linear equations derived from setting partial derivatives of MSE w.r.t each parameter to zero.
25. **Q:** Optimization of parameters for complex models is typically done numerically using what class of algorithms?
A: Iterative numerical optimization algorithms.
26. **Q:** What is the most commonly used numerical optimization algorithm for training ML model parameters, also known as the steepest descent algorithm?
A: Gradient Descent.
27. **Q:** In the Gradient Descent update rule, $\vec{\beta}^{(k+1)} = \vec{\beta}^{(k)} - \eta \nabla L(\vec{\beta}^{(k)})$, what does the scalar η (or η_0) represent, and why is it considered crucial?

A: η is the learning rate; it controls step size and convergence stability.

28. **Q:** When using Gradient Descent, the update vector $\Delta \vec{\beta}$ must point in what direction to ensure the highest magnitude of loss decrease?

A: Opposite to the gradient of the loss function.

29. **Q:** If all eigenvalues of the **Hessian matrix** (H) are **positive**, what is the resulting classification of the Hessian and the nature of the point considered?

A: Hessian is positive definite; the point is a local minimum.

30. **Q:** If the Hessian matrix (H) is **indefinite** (some eigenvalues positive, some negative), what is the resulting classification of the stationary point?

A: The point is a saddle point.